(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :09/03/2026

(21) Application No.202641027237 A

(43) Publication Date : 24/04/2026

(54) Title of the invention : EMBEDDED INTELLIGENCE BASED PREDICTIVE ENERGY MANAGEMENT AND ADAPTIVE CONTROL SYSTEM FOR THERMOELECTRIC WASTE HEAT RECOVERY IN INTERNAL COMBUSTION ENGINE VEHICLES.

(51) International classification	:F01N 5/02, H02N 11/00, H01L 35/30, F02G 5/02, F02G 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRI SAIRAM ENGINEERING COLLEGE(AUTONOMOUS)
Address of Applicant :SRI SAI RAM ENGINEERING COLLEGE, SAI LEO NAGAR, WEST TAMBARAM, CHENNAI-600044 Tamil Nadu India
(72)**Name of Inventor :**
1)SHANTHINI S
2)SUJITHRA R
3)BOOMIGA E R
4)Dr.ALAGU S

(57) Abstract :
The present invention relates to an embedded intelligence-based adaptive energy management system for thermoelectric waste heat recovery in internal combustion engine vehicles. The system is integrated with a thermoelectric generator (TEG) array configured to convert vehicular waste heat into electrical energy using the Seebeck effect. The invention comprises thermal sensing means, electrical sensing means. a DC-DC boost conversion module, and an embedded control unit configured to predict short-horizon thermal gradient behavior. The embedded control unit computes a thermally derived decision parameter termed Thermal Availability Index (TAl) based on temperature gradient magnitude, rate of change, and stability duration. The TAl is utilized to dynamically regulate boost conversion characteristics, perform module-level thermal equalization, and intelligently route energy towarC: battery storage or auxiliary loads. The system further includes a fault prognostic mechanism for detecting abnofiTlal Seebeck response patterns and voltage instability. The coordinated predictive regulation and adaptive control enhance electrical conversion efficiency, improve voltage stability, reduce thermal stress, and extend operational lifespan of thermoelectric modules under dynamically varying vehicular conditions.

No. of Pages : 25 No. of Claims : 10